

Module 13 - Type Conversion (Implicit & Explicit)

1. What is Type Conversion?

Type Conversion is the process of converting a value from one data type to another data type.

2. What is Implicit Type Conversion?

Implicit Type Conversion is the automatic conversion of a value from one data type to another by the programming language.

Example Code:

```
// Implicit Type Conversion
let str = "45";
let num = 100;
console.log(str + num);           // 45100
console.log(typeof(str + num));   // string

// String
console.log("Hi" + true);         // Hitrue
console.log("Hi" + undefined);    // Hiundefined
console.log("Hi" + null);         // Hinull
console.log("Hi" + [1,2]);        // Hi1,2
console.log("Hi" + {});           // Hi[object Object]

// String + Anything = String
// Anything + String = String

// Number
console.log(10 + true);           // 10 + 1 = 11
console.log(10 + false);          // 10 + 0 = 10
// true = 1, false = 0
console.log(10 + undefined);      // NaN - Not A Number
console.log(10 + null);           // 10 + 0 = 10
console.log(10 + [1,2]);          // 101, 2
console.log(typeof(10 + [1]));    // string
```

```
console.log(10 + {});           // 10[object Object]
console.log(typeof(10 + ""));   // string
console.log(10 - 'abc');        // NaN
console.log(10 - "");           // 10

// Boolean
console.log(true + 10);         // 11
console.log(true + undefined);  // NaN
console.log(true + null);       // 1
```

3. What is Explicit Type Conversion?

Explicit Type Conversion is the process of manually converting a value from one data type to another using a specific method or function.

Example Code:

```
// Explicit Type Conversion

// Number
console.log(10 + Number("10")); // 20
console.log(Number("abc"));      // NaN
console.log(Number(true));       // 1
console.log(Number(false));      // 0
console.log(Number([]));         // 0
console.log(Number([1]));        // 1
console.log(Number([1,2]));      // NaN
console.log(Number({}));         // NaN

// Boolean
console.log(Boolean(""));        // false
console.log(Boolean("123"));     // true
console.log(Boolean(10));        // true
console.log(Boolean(-10));       // true
console.log(Boolean(0));         // false
console.log(Boolean(undefined)); // false
```

```
console.log(Boolean(null));           // false
console.log(Boolean([]));             // true
console.log(Boolean([1]));            // true
console.log(Boolean({}));             // true
console.log(Boolean(Infinity));       // true
console.log(Boolean(-Infinity));      // true
console.log(Boolean(NaN));            // false

// String
console.log(String());               // ""
console.log(String(10));              // "10"
console.log(String(true));            // "true"
console.log(String(false));           // "false"
console.log(String(null));            // "null"
console.log(String(undefined));       // "undefined"
console.log(String([1,2]));           // "1,2"
console.log(String({}));              // "[object Object]"
```